

Distributed Elevator Control System Using SocketCAN

TEST AND VERIFICATION REPORT

1. Test Objective

The objective of testing was to verify the communication and operation of the distributed elevator control system using Linux SocketCAN and the virtual CAN interface `vcan0`.

The testing focused on verifying:

- Floor request handling
- Elevator movement
- Motor control
- Door control
- Position monitoring
- ECU heartbeat monitoring
- Fault detection and recovery
- HMI status reporting

2. Test Environment

Parameter	Configuration
Operating System	Ubuntu Linux
Programming Language	Python 3
CAN Interface	SocketCAN <code>vcan0</code>
Number of Floors	5
ECUs	6
HMI	Console-based

Each ECU was executed as an independent Python process. The ECUs communicated through CAN messages over the shared `vcan0` interface.

3. Test Cases

3.1 Floor Request Test

Test Description

A destination floor was entered through the Passenger Request ECU. The Passenger Request ECU transmitted the requested floor over CAN using the configured floor request message.

Expected Result

The Controller ECU should receive the requested floor and begin the elevator control sequence.

Actual Result

The requested floor was successfully transmitted and received by the Controller ECU.

Result : PASS

3.2 Elevator Movement Test

Test Description

After receiving a valid floor request, the Controller ECU determined the required direction and sent an appropriate motor command. The Position/Sensor ECU simulated elevator movement and transmitted the current floor periodically.

Expected Result

The elevator should move in the required direction and the current floor should update until the requested floor is reached.

Actual Result

The Position/Sensor ECU updated the current floor while the motor was commanded to move.

Result : PASS

3.3 Motor Control Test

Test Description

The Motor ECU was tested for the following commands: UP ; DOWN ; STOP

Expected Result

The Motor ECU should receive the command and report the corresponding motor status.

Actual Result

The Motor ECU correctly received the motor commands and reported the motor status.

Result : PASS

3.4 Door Control Test

Test Description

After reaching the requested floor, the Controller ECU sent commands to the Door ECU. The door was tested for opening and closing operation.

Expected Result

The door should change between the required states and report its current status.

Actual Result

The Door ECU correctly responded to the open and close commands.

The observed sequence was:

CLOSED → OPENING → OPEN

OPEN → CLOSING → CLOSED

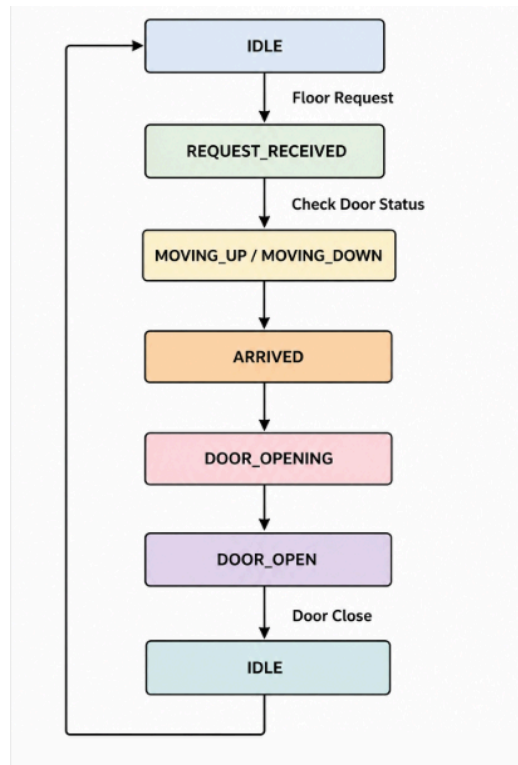
Result : PASS

3.5 Normal Operation Test

Test Description

A complete elevator request was tested from the initial idle state until the elevator returned to the idle state.

The normal sequence was:



Expected Result

The complete sequence should execute without a system fault.

Actual Result

The elevator successfully processed the floor request, moved to the requested floor, stopped the motor, operated the door and returned to the idle state.

Result : PASS

3.6 ECU Heartbeat Monitoring Test

Test Description

Each ECU periodically transmitted a heartbeat message. The Diagnostic ECU monitored these heartbeat messages and detected communication timeout when an ECU stopped transmitting.

Expected Result

The Diagnostic ECU should report a fault when the heartbeat is lost and report recovery when communication is restored.

Actual Result

The Diagnostic ECU successfully detected heartbeat timeout and later detected heartbeat recovery.

Example:

FAULT: Position/Sensor ECU heartbeat timeout
RECOVERY: Position/Sensor ECU heartbeat restored
SYSTEM RECOVERY: All monitored ECUs operational

Result : PASS

3.7 Invalid Position Test

Test Description

An invalid position status was used to verify the diagnostic monitoring functionality.

Expected Result

The Diagnostic ECU should detect the invalid position status and report a corresponding fault.

Actual Result

The Diagnostic ECU reported:

FAULT: Invalid position status

The HMI also displayed:

System Fault : INVALID POSITION
Last Event : INVALID POSITION

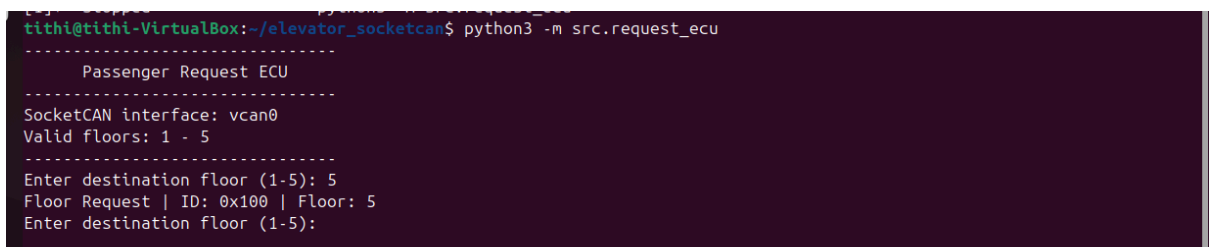
Result : PASS

4. Screenshots

The following screenshots were selected as evidence of the main system functions.

Screenshot 1 – Floor Request

Passenger Request ECU showing the destination floor being entered and transmitted.



```
tithi@tithi-VirtualBox:~/elevator_socketcan$ python3 -m src.request_ecu
-----
Passenger Request ECU
-----
SocketCAN interface: vcan0
Valid floors: 1 - 5
-----
Enter destination floor (1-5): 5
Floor Request | ID: 0x100 | Floor: 5
Enter destination floor (1-5):
```

Screenshot 2 – Elevator Movement

Controller and Position/Sensor ECU showing the motor command and changing floor position.

Position Sensor:

```
tithi@tithi-VirtualBox:~/elevator_socketcan$ python3 -m src.position_sensor_ecu
-----
Position/Sensor ECU
-----
SocketCAN interface: vcan0
Initial floor: 1
-----
Position | Floor: 2 | Direction: UP
Position | Floor: 3 | Direction: UP
Position | Floor: 4 | Direction: UP
Position | Floor: 5 | Direction: UP
```

Controller :

```
tithi@tithi-VirtualBox:~/elevator_socketcan$ python3 -m src.controller_ecu
-----
Controller ECU
-----
SocketCAN interface: vcan0
-----
Floor Request Received | Target: 5
Motor Command | ID: 0x200 | UP
Current Floor | 2
Current Floor | 3
Current Floor | 4
Current Floor | 5
Motor Command | ID: 0x200 | STOP
Door Command | ID: 0x201 | OPEN
Door Command | ID: 0x201 | CLOSE
SAFETY RECOVERY RECEIVED -> NORMAL OPERATION
Motor Command | ID: 0x200 | STOP
SAFETY COMMAND RECEIVED -> MOTOR STOP
SAFETY RECOVERY RECEIVED -> NORMAL OPERATION
```

Motor :

```
tithi@tithi-VirtualBox:~/elevator_socketcan$ python3 -m src.motor_ecu
-----
Motor ECU
-----
SocketCAN interface: vcan0
Initial state: STOPPED
-----
Motor Command | UP
Motor Command | STOP
Motor Command | STOP
```

Screenshot 3 – Door Operation

Door ECU showing the opening and closing operation.

```
tithi@tithi-VirtualBox:~/elevator_socketcan$ python3 -m src.door_ecu
-----
Door ECU
-----
SocketCAN interface: vcan0
Initial state: CLOSED
-----
Door Command | OPEN
Door Status | OPEN
Door Command | CLOSE
Door Status | CLOSED
```

Screenshot 4 – HMI System Status

Console-based HMI showing the current floor, requested floor, controller state, actuator status and ECU status.

```
tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x
SYSTEM STATUS
-----
Current Floor      : 5
Requested Floor    : 5
Controller State   : IDLE
Position Status    : VALID

ACTUATORS
-----
Motor Status       : STOPPED
Door Status        : CLOSED

ECU STATUS
-----
Passenger Request  : ONLINE
Position/Sensor    : ONLINE
Controller         : ONLINE
Motor              : ONLINE
Door               : ONLINE

DIAGNOSTICS
-----
System Fault       : NONE
Last Event         : System normal

=====
SocketCAN interface: vcan0
=====
```

Screenshot 5 – Diagnostic Monitoring

Diagnostic ECU showing heartbeat fault detection and recovery.

```
tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x tithi@tithi... x
2026-08-23 16:36:31 | RECOVERY: Passenger Request ECU heartbeat restored
2026-08-23 16:36:36 | FAULT: Passenger Request ECU heartbeat timeout
2026-08-23 16:36:36 | FAULT: Position/Sensor ECU heartbeat timeout
2026-08-23 16:36:36 | FAULT: Controller ECU heartbeat timeout
2026-08-23 16:36:36 | FAULT: Motor ECU heartbeat timeout
2026-08-23 16:36:36 | FAULT: Door ECU heartbeat timeout
2026-08-23 16:36:36 | RECOVERY: Passenger Request ECU heartbeat restored
2026-08-23 16:36:36 | RECOVERY: Position/Sensor ECU heartbeat restored
2026-08-23 16:36:36 | RECOVERY: Controller ECU heartbeat restored
2026-08-23 16:36:36 | RECOVERY: Motor ECU heartbeat restored
2026-08-23 16:36:36 | RECOVERY: Door ECU heartbeat restored
^C
Diagnostic ECU stopped.
2026-08-23 16:38:42 | Diagnostic ECU stopped
tithi@tithi-VirtualBox:~/elevator_socketcan$ python3 -m src.diagnostic_ecu
-----
Diagnostic ECU
-----
SocketCAN interface: vcan0
Heartbeat timeout: 500 ms
Startup grace period: 2 s
-----
2026-08-23 16:40:17 | Diagnostic ECU started
2026-08-23 16:41:14 | FAULT: Controller ECU heartbeat timeout
2026-08-23 16:41:15 | RECOVERY: Controller ECU heartbeat restored
2026-08-23 16:41:15 | SYSTEM RECOVERY: All monitored ECUs operational
2026-08-23 16:42:05 | FAULT: Position/Sensor ECU heartbeat timeout
2026-08-23 16:42:05 | RECOVERY: Position/Sensor ECU heartbeat restored
2026-08-23 16:42:05 | SYSTEM RECOVERY: All monitored ECUs operational
```

5. Results

Test Case	Result
Floor Request	PASS
Elevator Movement	PASS
Motor Control	PASS
Door Control	PASS
Normal Operation	PASS
ECU Heartbeat Monitoring	PASS
Invalid Position Detection	PASS

The tests confirmed that the major functions of the distributed elevator system operated correctly over the SocketCAN `vcan0` interface.

6. Observations

The following observations were made during testing:

1. The Passenger Request ECU successfully transmitted valid floor requests through CAN.
2. The Controller ECU correctly processed the requested floor and generated motor and door commands.
3. The Position/Sensor ECU successfully simulated elevator movement and provided current floor information.
4. The Motor ECU correctly responded to `UP`, `DOWN` and `STOP` commands.
5. The Door ECU correctly responded to open and close commands and reported its current status.
6. The Diagnostic ECU was able to detect missing ECU heartbeats and identify when communication was restored.
7. Invalid position information was detected and reported by the diagnostic system.
8. The HMI provided a consolidated view of the elevator state, actuator status, ECU status and diagnostic information.

7. Conclusion

The testing successfully verified the main functions of the distributed elevator control system.

The individual ECUs communicated through the SocketCAN `vcan0` interface and performed their assigned functions. Floor requests, elevator movement, motor control, door operation and position monitoring were successfully demonstrated.

The diagnostic functionality was also verified through heartbeat timeout, heartbeat recovery and invalid position detection.

The test results show that the implemented system provides a working software-based demonstration of a distributed elevator control system using SocketCAN and Python.